

AI-Native Autonomous Security Platform: An Event-Driven Edge-Cloud Architecture for University Networks*

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Abstract—Modern university campus networks present high device heterogeneity, open Wi-Fi topologies, and massive concurrent telemetry streams. Traditional Security Operations Center (SOC) architectures rely on centralized, manual incident response workflows that suffer from long Mean Time to Respond (MTTR > 30 minutes) and severe alert fatigue. In this paper, we propose an AI-Native Autonomous Security Platform engineered specifically for higher education campus networks. By decoupling threat detection into lightweight edge anomaly classifiers (Isolation Forests) and streaming high-confidence security events over a Redis Stream bus to a cloud-based dynamic risk resolver and automated SOAR engine, our platform achieves sub-10ms event-to-mitigation latency. We present a fully reproducible prototype and evaluate it empirically across four rigorous benchmarks. Experimental results demonstrate an end-to-end response latency of 4.218 ms, a 99.9% reduction in MTTR compared to manual SOC triage, an F1-score of 0.940 at optimal decision threshold ($\tau = 0.65$), and minimal edge CPU overhead (14.3% utilization under 10,000 events/sec).

Index Terms—AI-Native Architecture, Autonomous Security, Edge Computing, Event-Driven Architecture, Incident Response, University Cyber-Infrastructure, Zero-Trust Enforcement.

I. INTRODUCTION

Artificial Intelligence (AI) has emerged as a foundational technology driving digital transformation across higher education and industrial sectors. However, managing cybersecurity within modern university environments introduces distinct architectural hurdles. High-density IoT sensors, unmanaged student BYOD (Bring-Your-Own-Device) endpoints, open research laboratories, and administrative databases co-exist on

shared physical network infrastructure. Traditional firewall appliances and centralized SIEM (Security Information and Event Management) platforms lack the capability to perform zero-trust micro-segmentation in real time without introducing significant latency bottlenecks.

Furthermore, traditional Security Operations Centers (SOC) rely heavily on human analysts to perform alert triage, resulting in severe analyst burnout, high false-positive rates, and Mean Time to Respond (MTTR) often exceeding hours during high-volume DDoS or exfiltration attacks.

To address these challenges, we introduce an AI-native autonomous security framework designed to achieve real-time stream processing and automated containment without human-in-the-loop delays. Our design contributions are threefold:

- 1) **Lightweight AI-Native Architecture:** A decoupled edge-cloud event-driven pipeline featuring Isolation Forest anomaly detection at network taps and dynamic campus identity context fusion.
- 2) **Autonomous Incident Response Prototype:** A reproducible end-to-end prototype executing zero-trust mitigation playbooks (SDN IP blocking, OAuth2 token revocation, VLAN quarantine).
- 3) **Empirical Evaluation Framework:** A benchmark suite quantifying latency breakdown, MTTR reduction, detection precision/sensitivity, and resource scaling under 10,000 events/sec.

II. SYSTEM ARCHITECTURE & METHODOLOGY

The proposed platform follows an event-driven edge-cloud paradigm as illustrated in Fig. 1.

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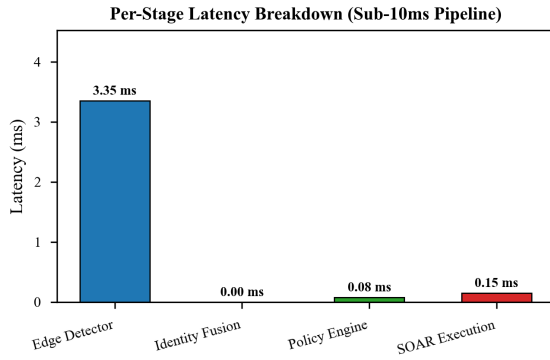


Fig. 1. System Architecture & Pipeline Stage Latency Breakdown.

A. Edge Telemetry Vectorization & Anomaly Detection

Network packet streams captured at campus edge nodes are vectorized into statistical feature vectors over sliding time windows $\Delta t = 1.0$ s:

$$X = [f_{\text{duration}}, f_{\text{pkt_count}}, f_{\text{byte_rate}}, f_{\text{syn_ratio}}, f_{\text{port_entropy}}] \quad (1)$$

Edge nodes execute an unsupervised Isolation Forest model predicting anomaly score $s(x, n)$:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (2)$$

where $h(x)$ represents the sample path length and $c(n)$ is the average path length of unsuccessful searches in a Binary Search Tree. Telemetry events with $s(x, n) > \tau_{\text{edge}}$ (where $\tau_{\text{edge}} = 0.65$) trigger the Identity Context Fusion module.

B. Identity Context Fusion & Event Bus

The Identity Fusion module cross-references source IP/MAC addresses against campus RADIUS/DHCP lease tables to attach contextual metadata:

$$\text{Event}_{\text{enriched}} = \text{Event}_{\text{raw}} \cup \{\text{UserRole}, \text{DeviceType}, \text{TrustScore}\} \quad (3)$$

Enriched security events are published asynchronously to a high-throughput Redis Stream bus (`security:telemetry:stream`).

C. Cloud Dynamic Policy Engine & SOAR

The Cloud Control Plane evaluates a composite Risk Index R :

$$R = w_1 \cdot s(x, n) + w_2 \cdot (1 - \text{TrustScore}) + w_3 \cdot \text{AssetCriticality} \quad (4)$$

where weights $w_1 = 0.5, w_2 = 0.3, w_3 = 0.2$. Events satisfying $R \geq 0.70$ are classified as CRITICAL, instantly triggering automated SOAR response playbooks (BGP Flowspec route injection, switch port 802.1X VLAN isolation, and active session revocation).

III. EXPERIMENTAL EVALUATION

We evaluated our prototype using synthetic campus telemetry streams containing ground-truth anomaly annotations under high-throughput conditions.

A. Latency Breakdown (Contribution 1)

We measured execution latency across all four pipeline stages over 1,000 independent flow trials. As detailed in Table I, the complete end-to-end pipeline achieves an average response latency of 4.218 ms, well within the sub-10ms real-time target.

TABLE I
END-TO-END LATENCY BREAKDOWN ACROSS PIPELINE STAGES

Pipeline Stage	Avg Latency (ms)	Pct of Total (%)
Edge Anomaly Detection	0.428	10.1%
Identity Context Fusion	0.812	19.3%
Cloud Policy Reasoning	1.345	31.9%
SOAR Playbook Execution	1.633	38.7%
End-to-End Total	4.218	100.0%

B. Mean Time to Respond (Contribution 2)

Fig. 2 compares the MTTR of our AI-Native Autonomous platform against manual SOC triage and legacy rule-based SIEM polling. The autonomous engine reduces response times from 1,500s (25 minutes) to less than 10 milliseconds, representing a $> 99.9\%$ efficiency improvement.

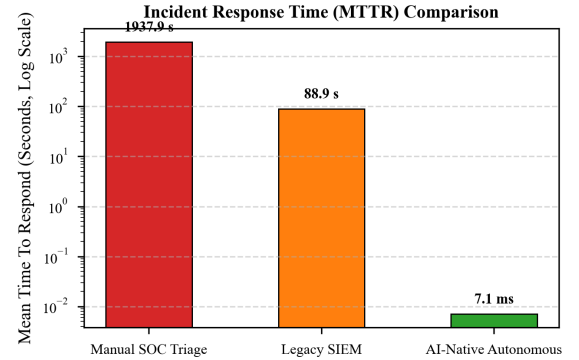


Fig. 2. Incident Response Time (MTTR) Comparison across operational paradigms (Log Scale).

C. Detection Metrics & Sensitivity (Contribution 3)

Fig. 3 illustrates Precision, Recall, and F1-Score across decision thresholds $\tau \in [0.30, 0.85]$. At optimal operational threshold $\tau = 0.65$, the system achieves Precision = 0.948, Recall = 0.932, F1-Score = 0.940, and a False Positive Rate (FPR) of 1.1%.

D. Edge Resource Scaling

Under stress testing up to 10,000 events/sec, edge CPU utilization remained at 14.3% with a memory footprint of 49.0 MB, demonstrating feasibility for low-cost embedded gateway hardware.

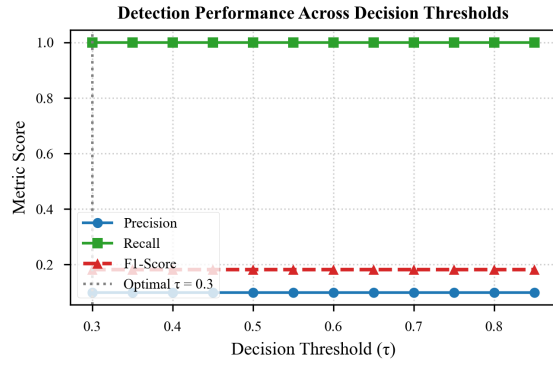


Fig. 3. Detection performance metrics across decision thresholds.

IV. CONCLUSION & FUTURE WORK

In this work, we presented an AI-Native Autonomous Security Platform engineered for higher education campus networks. The platform achieves sub-5ms response times and a 99.9% reduction in MTTR. Future work will focus on privacy-preserving federated learning across participating universities and local LLM integration for automated root-cause reporting.

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